A 28nm 44.15TFLOPS/W and 1.57TFLOPS/mm² Joint-Alignment Floating-point CIM Macro with ISO26262 ASIL-D Safety Level for Autonomous Driving

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Abstract

This paper presents a floating-point Computing-in-Memory (FP-CIM) macro tailored for AI model acceleration in autonomous driving. It has three features: 1) A joint-alignment CIM structure that performs the exponent-mantissa joint-alignment, reducing 3.97x accuracy loss and 31.24% area efficiency loss. 2) A hybrid XNOR/XOR adder tree (HAT) featuring first-level XNOR-based 6T and other-level XOR-based 10T full adders (FA), reducing 71.35% area overhead. 3) A unified all-inclusive Silent Data Corruption (SDC) decoder (UASD) that directly protects final MAC results against SDC, achieving <0.5% SDC rate with only 8.41% area cost and ISO 26262 ASIL-D safety level.

Introduction

Artificial intelligence (AI) is revolutionizing autonomous driving, delivering exceptional performance in safety-critical tasks like obstacle avoidance and path planning. SRAM-based FP-CIMs [1-5] offer superior computing density, energy efficiency, and precision, making them promising to accelerate these AI models in high-performance edge scenarios. However, autonomous driving imposes stricter demands on accuracy, performance, and reliability of hardware accelerators [7]. Deploying FP-CIM in such contexts faces three key challenges (Fig.1&2): 1) Prior FP-CIM designs rely on exponent-only alignment and extend mantissa bit-width to reduce truncation accuracy loss, but this extension introduces 22-44% extra compute latency in bit-serial CIM. 2) The large area overhead of adder trees (34–71% of CIM macro) leads to low area efficiency and restricts overall performance, hindering its ability to meet the demand for high-performance computation. 3) SDC [8] from soft errors or manufacturing defects, threatens memory and compute logic reliability in CIM, posing safety risks to autonomous driving. Traditional error correction codes (ECC) [9-10] only protect memory, necessitating separate triple modular redundancy (TMR) [11] for compute logic. However, such separate protection incurs prohibitive area and power cost, despite meeting ISO 26262 ASIL-D safety requirements [12].

Proposed FP-CIM Overall Architecture

We tackle these challenges by implementing an FP-CIM with three key features: 1) A joint-alignment CIM structure that performs joint-alignment, which repurposes the otherwise wasted bit-space (i.e., inherent-invalid bits) between the hidden bit (H) and mantissa's leading-one to accommodate more effective bits, minimizing truncation loss without bit-width extension. 2) The HAT featuring first-level XNOR-based 6T and other-level XOR-based 10T FAs, which precomputes and reuses the common XOR subexpression shared between Sum and carry-out (Cout), reducing FA transistor count. 3) The UASD directly protects final MAC results against SDC. Weight data are encoded with residue codes and prestored in CIM. UASD detects MAC errors via remainder checking and restore correct results through the unique remainder-residual mapping, achieving the ISO 26262 ASIL-D safety level. Fig. 3 presents the overall architecture. The input joint-alignment unit (IJAU) first performs the exponent-mantissa joint-alignment for the FP inputs, which are then sent through the input selector (IS) to CIM subarrays for the bit-wise multiplication and hidden-bit compensation via a hidden-bit compensator (HC). The HAT accumulates partial products from 32 subarrays. MAC results are sent to the UASD for correctness verification by modulus operations. Upon detection of SDC, the remainder is served as an index to residual LUT, restoring the correct MAC result.

Fig. 4&5 depict the structure of the **joint-alignment CIM**, performing exponent-mantissa joint-alignment via the IJAU, IS, and HC components. This process includes three steps: **1) Mantissa leading-one alignment**. The IJAU separates H and mantissa for each FP input. The leading-one position detector (LPD) loads 32 input mantissas and detects their leading-one positions (LP). Based on LPs, the mantissa shifter left-shifts each mantissa accordingly to fully utilize the preceding inherent-invalid bits. **2) Exponent alignment.** The exponent normalizer updates exponents (UE_i) based on their LPs, identifies the maximum updated exponent (UE_{Max}), and calculates exponent offsets (ΔUE_i). The mantissa shifter then is reused to right-shift leading-one mantissas by respective ΔUE_i. **3) Hidden-bit compensation.** Aligned mantissas are used for MAC operations, where H compensation operates in two modes, determined by hidden-bit positions (HP). For in-window mode, when the bit-serial input counter matches HP, IS sets the input to 1 for compensation; For out-window mode, the HC activates Compensation Flag and left-shifts entire row data by LP, finishing compensation in one cycle. Our design reduces 3.39× SER [13] and 3.97× accuracy loss over [3], while avoiding 31.24% area efficiency loss over [6] with 1.62% extra area cost.

Fig. 6&7 depict the structure of the **HAT**. Unlike 28T adder trees [14], our HAT employs transmission-gate logic with only 6T and 10T FAs for rail-to-rail operation. The key is to pre-compute and reuse the common XOR term $P = A \oplus B$ shared in Sum and Cout, reducing transistor count. Considering the reusability of bit-wise multiplication results, we adopt a bipolar format (BF) for mantissas, using a full-swing 6T XNOR gate as the bit-wise multiplier. Each BF bit represents ±1 and maps losslessly to sign-magnitude [13]. **1) First-level FAs** compute the Sum and Cout of $\overline{(I_0 \oplus A)}$ and $\overline{(I_1 \oplus B)}$, where the shared logic is $P = \overline{(A \oplus B)} \oplus (I_0 \oplus I_1)$. By exploiting the inherent XNOR logic of the multiplier, the shared P is pre-computed to further reduce transistor count. Specifically, one CIM subarray stores A (W_0) and computes $R_0 = I_0 \oplus A$, while another stores the pre-computed $A \oplus B$ ($W_0 \oplus W_1$) directly within subarrays instead of storing B separately. $I_0 \oplus I_1$ is then applied as CIM input to complete the P computation. This design enables the first-level FA to operate with only 6 transistors while maintaining signal integrity. **2) Other-level FAs** that perform A+B summation, employ a 4T full-swing XOR gate to pre-compute the shared term $P = A \oplus B$, enabling rail-to-rail computation with only 10 transistors. Our HAT reduces 71.35% area cost over the 28T adder tree [14], achieving average 1.54x area efficiency gain.

Fig. 8&9 present the structure of the **UASD**. Unlike protecting memory and compute logic separately, our unified protection strategy offers robust SDC correction capability, enabling direct correction for the MAC error by subtracting a unique residual corresponding to its remainder. Each row stores a 137b Codeweight, including a 128b weight operand and 9b residues pre-encoded with residue codes, which is an integer multiple of a prime N. The UASD comprises a MAC result concatenator, an SDC detector&corrector, and a macro-shared 128×9b residual LUT. After MAC operations, the concatenator merges 17 results (16 weight and one residue results) according to their encoding positions. The CodeMAC is sent to the detector for a modulo-N check. A non-zero remainder indicates an SDC, which is served as an index to the residual LUT, restoring the correct MAC results by its unique residual. To minimize latency, we design a fast modulo unit that computes the remainder directly from fractional bits discarded in conventional modulo operations, alongside a Radix-4 Booth encoding constant multiplier, reducing partial products by half and modulo latency by 24.29%. The proposed macro achieves an average 0.50% SDC rate, 28.69× lower than unprotected CIMs, meeting ASIL-D safety level. Unified protection incurs only 8.41% area and 8.15% power cost, saving 48.44% area and 60.09% power over separate protection with same safety level.

Measurement Results

The PF-CIM test chip is fabricated in 28nm with 1.92 mm² area (Fig. 10). The highest energy efficiency is 44.15TFLOPS/W for BF16 and 59.16TOPS/W for INT8 at 0.62V, 150MHz with 1.51-1.62× improvement over the prior work [3] (Table. I). The joint-alignment maintains average 0.88% accuracy loss on typical AI models with 3.39x SER improvement over prior FP-CIMs [3, 6]. Thanks to HATs, our CIM macro achieves area efficiency of 1.57TFLOPS/mm² for BF16 (15.10× over [6]; 1.37× over [3]) and 1.77TOPS/mm² for INT8 (2.42× over [10]; 1.24× over [3]) at 1.0V, 500MHz. Compared to memory-only protection CIMs [9, 10], our unified protection achieves 99.50% fault coverage with ISO 26262 ASIL-D safety level.

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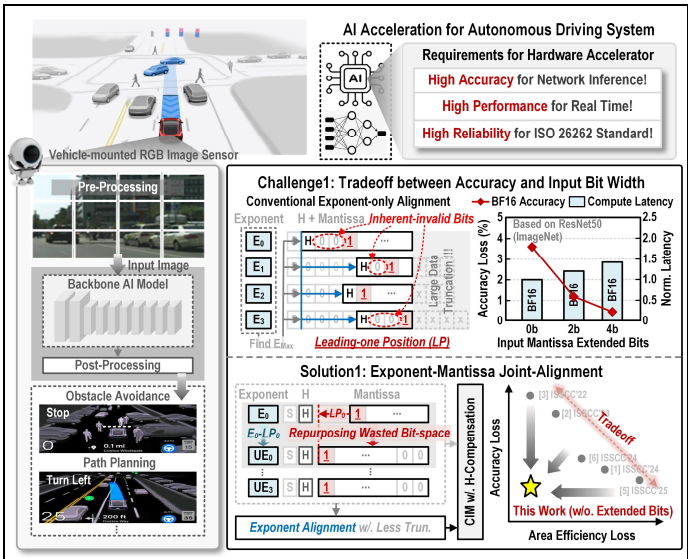


Fig. 1. Requirements for hardware accelerators, 1st challenge of deploying FP-CIM for AI model acceleration in autonomous driving, and proposed Solution1 (Exponent-Mantissa Joint-Alignment).

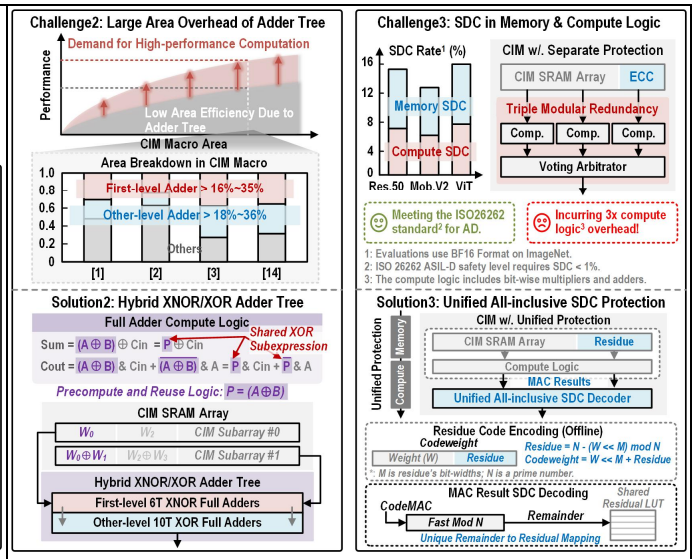


Fig. 2. 2nd and 3rd challenges of deploying FP-CIM in autonomous driving, and proposed Solution2 (Hybrid XNOR/XOR Adder Tree) and Solution3 (Unified All-inclusive SDC Protection).

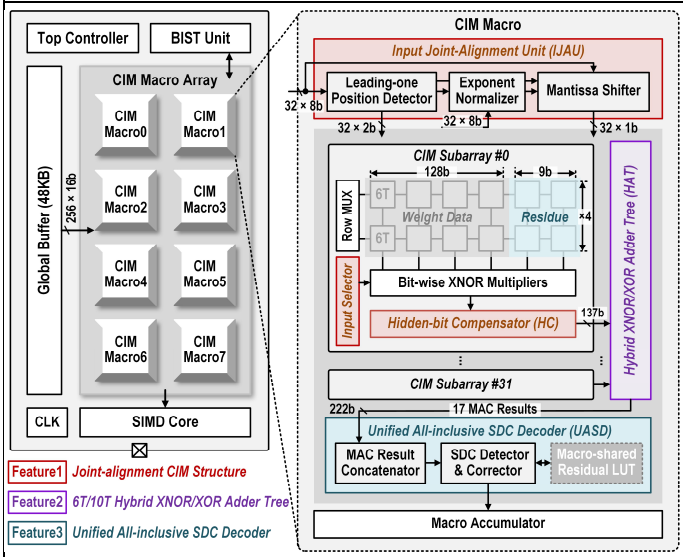


Fig. 3. Overall architecture of the proposed FP-CIM and its three key features.

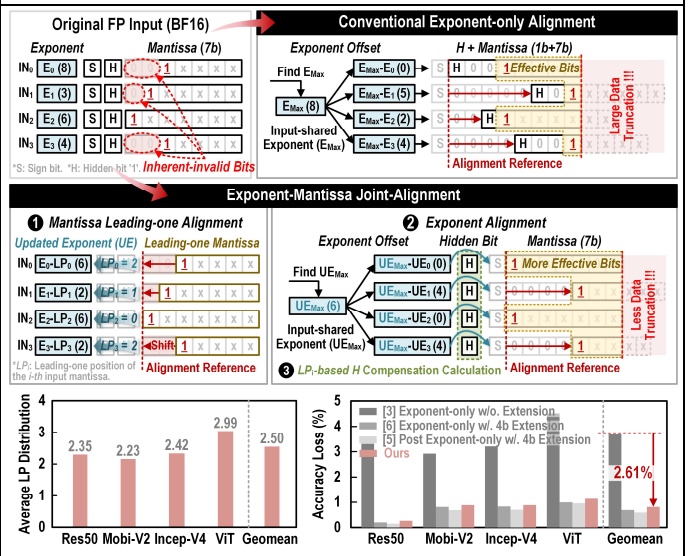


Fig. 4. Exponent-mantissa joint-alignment method and accuracy evaluation, where the joint-alignment includes mantissa leading-one alignment, exponent alignment, and hidden-bit compensation.

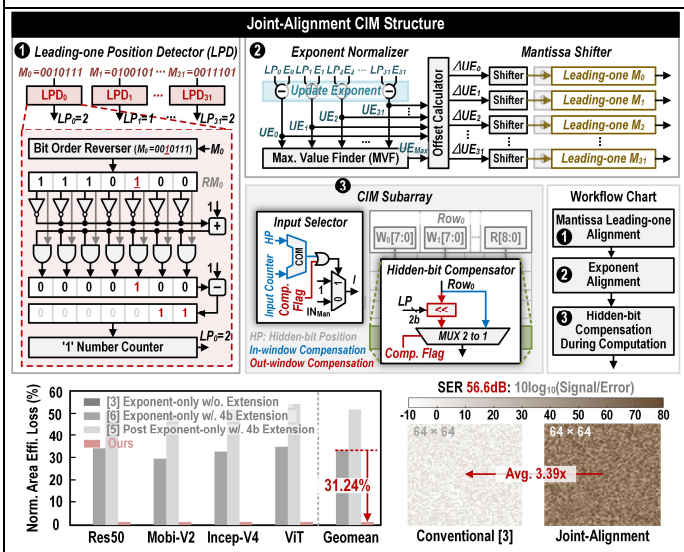


Fig. 5. Joint-alignment CIM structure and its detailed implementation for exponent-mantissa joint-alignment, and its area efficiency and signal error rate (SER) evaluation.

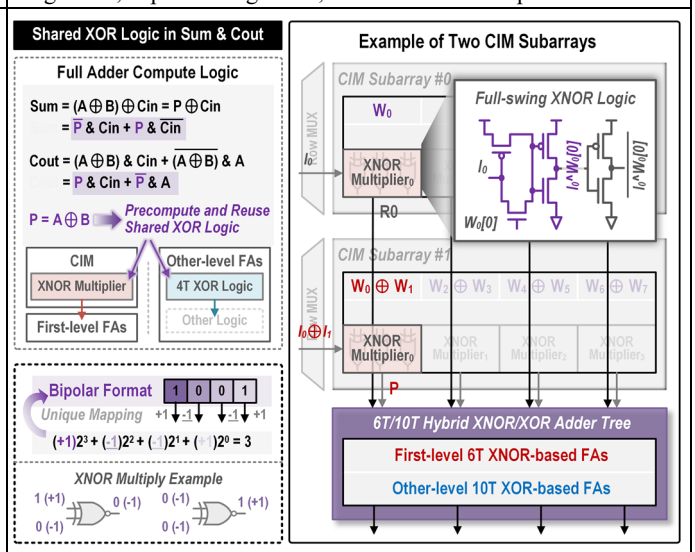


Fig. 6. Principle of hybrid XNOR/XOR adder tree, bipolar format, and accumulation example of two CIM subarrays.

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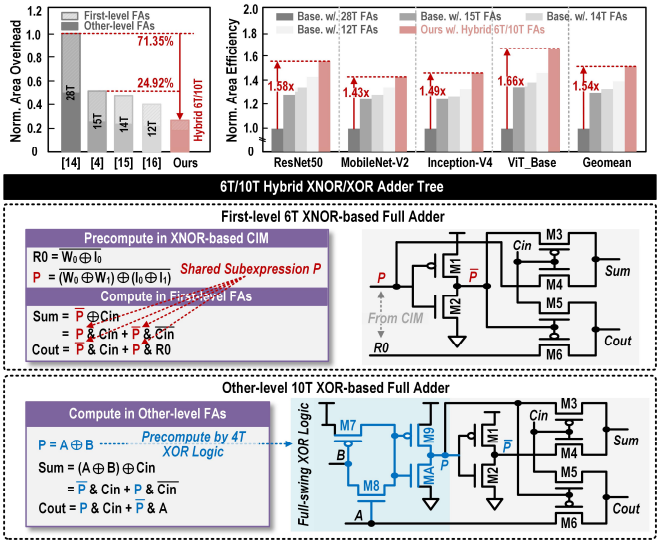


Fig. 7. Structure of hybrid XNOR/XOR adder tree featuring first-level 6T XNOR-based full adder and other-level 10T XOR-based full adder, and its evaluation.

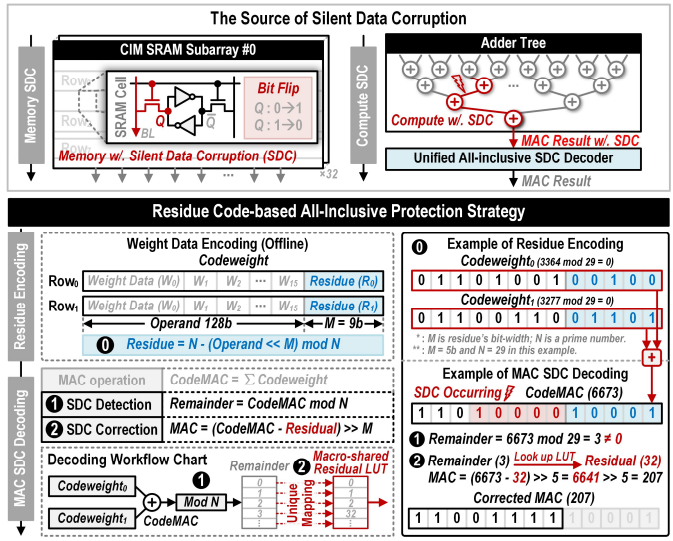
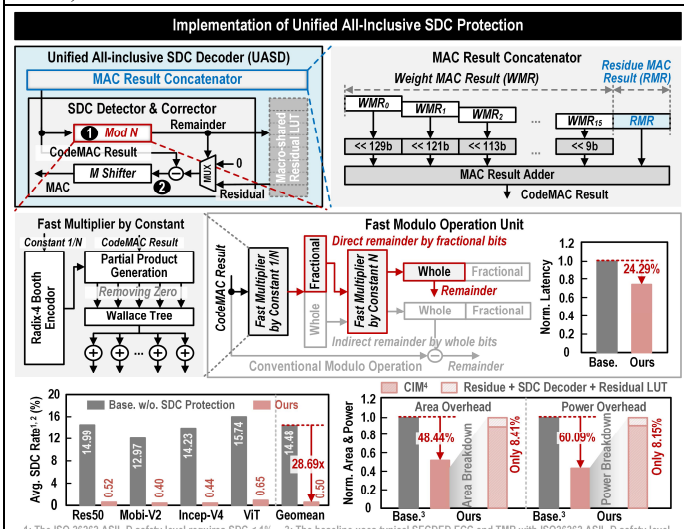


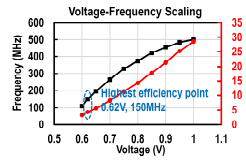
Fig. 8. Source of silent data corruption in CIM macro, proposed residue code-based all-inclusive protection strategy for final MAC results, and example of encoding, detection and correction.

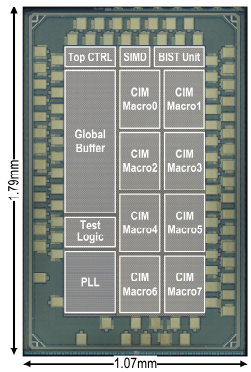


1: The ISO 26262 ASIL-D safety level requires SDC < 1%. 3: The baseline uses typical SECDED ECC and TMR with ISO26262 ASIL-D safety level.
2: Evaluations use BF16 format based on ImageNet. 4: It represents CIM logic used for weight, excluding SDC protection-related overhead.

Fig. 9. Structure of unified all-inclusive SDC decoder and its detailed implementation, achieving average 0.50% SDC rate and ISO 26262 ASIL-D safety level, with only 8.41% area and 8.15% power cost.

Model	ResNet50		MobileNet-V2		ViT_Base	
Dataset	ImageNet		ImageNet		ImageNet	
Data Precision	INT8	BF16	INT8	BF16	INT8	BF16
Top-1 Accuracy	75.85%	76.09%	70.67%	71.59%	81.29%	81.73%
Accuracy Loss	0.0%	0.57%	0.0%	0.83%	0.0%	1.31%
Energy Efficiency[1] (TOPS/W or TFLOPS/W)	37.76	29.65	37.13	29.15	39.79	31.24
Execution Time[2] (ms)	17.32	19.48	23.91	26.90	70.15	78.91
Inferences/s[2] (FPS)	57.74	51.33	41.82	37.17	14.26	12.67
Avg. SDC Rate	0.23%	0.52%	0.22%	0.40%	0.37%	0.65%





Application	AI Acceleration in Autonomous Driving	
Technology	TSMC 28nm CMOS	
Die Area	1.79mm×1.07mm	
Supply Voltage	0.6-1.0V	
Frequency	110-500MHz	
Buffer Size	48KB	
CIM Size	137Kb	
Alignment	Exponent-mantissa Joint-alignment	
Accuracy	Signal Error Rate: 56.6dB	
	Avg. Accuracy Loss: 0.88%	
Power	3.25mW @0.60V, 110MHz	
	28.56mW @1.0V, 500MHz	
Peak Performance[1]	INT8	0.51TOPS
	BF16	0.46TFLOPS
Energy Efficiency[2]	INT8	Macro: 59.16 TOPS/W System: 40.95 TOPS/W
	BF16	Macro: 44.15 TFLOPS/W System: 32.15 TFLOPS/W
Area Efficiency[1]	INT8	Macro: 1.77 TOPS/mm² System: 0.27 TFLOPS/mm²
	BF16	Macro: 1.57 TFLOPS/mm² System: 0.24 TFLOPS/mm²
SDC Rate	Avg. 0.50%	
ISO 26262 Assessment	ASIL-D Safety Level	

*1: Measured at the highest efficiency point, 0.62V, 150MHz. *2: Measured at the highest performance point, 1.0V, 500MHz.

Fig. 10. Measurement results for representative AI models, die photo, voltage-frequency scaling curves, and summary table.

TABLE I. Comparison with state-of-the-art CIM macros.

		ISSCC'24-34.8 [6]	ISSCC'23-29.5 [9]	ISSCC'22-15.5 [3]	JSSC'25 [10]	This Work
MAC Implementation		Analog CIM	Analog CIM	Digital CIM	Digital CIM	**Digital CIM**
Alignment Strategy		× Exponent-only Alignment	N/A	× Exponent-only Alignment	N/A	√ **Exponent-mantissa Joint-alignment**
SDC Protection		N/A	× Memory-only	N/A	× Memory-only	√ **Unified Memory-compute**
Data Precision		BF16, FP16	INT1/4/8	FP32, BF16, INT8/16	INT8/16	**BF16, INT8**
Technology		22nm	40nm	28nm	28nm	**28nm**
Supply Voltage (V)		0.7-0.8	0.9	0.6-1.0	0.55-1.0	**0.6-1.0**
Die Area (mm²)		8.20	20.25	6.69	3.03	**1.92**
Peak Performance[*1] (TOPS or TFLOPS)	INT8	-	0.23	1.35	0.17	**0.51**[*3]
	BF16	0.86	-	1.08	-	**0.46**[*3]
Energy Efficiency[*1] (TOPS/W or TFLPOS/W)	INT8	-	1.43	36.5	19.5	**59.16**[*2]
	BF16	65.5 (input sparsity 90%)	-	29.2	-	**44.15**[*2]
Area Efficiency[*1] (TOPS/mm² or TFLOPS/mm²)	INT8	-	0.011	1.43	0.73	**1.77**[*3]
	BF16	0.104	-	1.15	-	**1.57**[*3]
Average Fault Coverage[*4]		-	91.37%	-	92.35%	**99.50%**
ISO 26262 Assessment[*4]		N/A	**ASIL-B**	N/A	**ASIL-B**	**ASIL-D**

*1: Evaluated in the dense mode. One operation (OP) represents one multiplication or one addition.
*2: Measured at the highest efficiency point, 0.62V, 150MHz.
*3: Measured at the highest performance point, 1.0V, 500MHz.
*4: ISO26262 has a metric for autonomous driving reliability i.e., single-point fault coverage, where the requirements for ASIL are 90%, 97%, and 99% for B, C and D levels, respectively.

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